In [9]:

```python
import pandas as pd
arr=pd.Series([1,2,3,4,56],index=['A','B','C','D','E'])
print(arr)
print(arr['E'])
#vectorized operation no loop required
print(arr+5)
print(arr.iloc[1])
print(arr.loc['A'])
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())
print(arr.var())

#Boolean Indexing
print(arr[arr>30])

#count values
print(len(arr))
# it return indivitual value count
print(arr.value_counts())

#string operation
name = pd.Series(['kavya','mithun','riya'])
print(name.str.upper())
print(name.str.len())
```

```
A     1
B     2
C     3
D     4
E    56
dtype: int64
56
A     6
B     7
C     8
D     9
E    61
dtype: int64
2
1
66
23.95203540411545
56
1
13.2
3.0
573.6999999999999
E    56
dtype: int64
5
1     1
2     1
3     1
4     1
56    1
Name: count, dtype: int64
0     KAVYA
1    MITHUN
2      RIYA
dtype: object
0    5
1    6
2    4
dtype: int64
```

In [ ]:

In [7]:
```python
import pandas as pd
a = pd.Series(['divi','arthi','priya','moli'])
print(a)
print(a.values)
print(a.index)
print(a.dtype)

#accessing elements
print(a[1])

#accessing element using loc
print(a.loc[2])
print(a.iloc[2])
```

```
#slicing
print(a[1:3])
print(a.loc[1:3])
print(a.iloc[1:3])
```

```
0     divi
1    arthi
2    priya
3     moli
dtype: object
['divi' 'arthi' 'priya' 'moli']
RangeIndex(start=0, stop=4, step=1)
object
arthi
priya
priya
1    arthi
2    priya
dtype: object
1    arthi
2    priya
3     moli
dtype: object
1    arthi
2    priya
dtype: object
```

In [11]:
```
import pandas as pd
a = pd.Series({
    "gayu":16,
    "rithi":22,
    "ramya":28,})
print(a)
print(a["gayu"])
```

```
gayu     16
rithi    22
ramya    28
dtype: int64
16
```

In [12]:
```
#NaN - float type value
#index aligment
s1 = pd.Series([1,2,3,4,5])
s2 = pd.Series([1,2,3],index=['B','C','d'])
print(s1+s2)


s = pd.Series([1,2,3,4,5])
print(s.count())
print(s.unique())
print(s.nunique())
```

```
0   NaN
1   NaN
2   NaN
3   NaN
4   NaN
B   NaN
C   NaN
d   NaN
dtype: float64
5
[1 2 3 4 5]
5
```

data frame

In [35]:

```
import pandas as pd
#method 1 : From Dictionary
data = ({
    'name':['arthi','priya','divi'],
    'roll_no':[4,3,8],
    'age':[21,28,23,]})
df = pd.DataFrame(data)
print(df)

#method 2 : From series to DataFrame
s1 = pd.Series([1,2,3])
s2 = pd.Series([4.9,5.4,6.3])
s3 = pd.Series([True,False,True])
df = pd.DataFrame({'sno' : s1,
                   'cgpa' : s2,
                   'attendence' : s3})
print(df)
#list of list
data =[['dharshini',22,98],
    ['dhara',21,88],
    ['pavi',20,77]]
df=pd.DataFrame(data,columns=["name","age","mark"])
print(df)

#list of Dictionary
data = [
    {"name": "bhuvi","course": "cs","grade":"A"},
    {"name": "vino","course":"bca","grade":"A++"},
    {"name": "subha","course":"aiml","grade":"A"}]
df = pd.DataFrame(data,index = ["std1","std2","std3"])
print(df)
#Adding one extra columns
df['marks']=[99,66,56]
print(df)


#Adding multiple rows
std4 = pd.DataFrame([{"name":"kathir","course":"cs","grade":"A+" , "marks" :90}],in
df = pd.concat([df,std4])
print(df)
```

```
#create multiple rows
students = pd.DataFrame([{"name":"rani","course":"ds","grade":"A+","marks": 90},
                         {"name":"thara","course":"csai","grade":"A","marks":99}],i
df = pd.concat([df,students])
print(df)
```

```
    name  roll_no  age
0  arthi        4   21
1  priya        3   28
2   divi        8   23
   sno  cgpa  attendence
0    1   4.9        True
1    2   5.4       False
2    3   6.3        True
        name  age  mark
0  dharshini   22    98
1      dhara   21    88
2       pavi   20    77
       name course grade
std1  bhuvi     cs     A
std2   vino    bca    A++
std3  subha   aiml     A
       name course grade  marks
std1  bhuvi     cs     A     99
std2   vino    bca    A++     66
std3  subha   aiml     A     56
        name course grade  marks
std1   bhuvi     cs     A     99
std2    vino    bca    A++     66
std3   subha   aiml     A     56
std4  kathir     cs    A+     90
        name course grade  marks
std1   bhuvi     cs     A     99
std2    vino    bca    A++     66
std3   subha   aiml     A     56
std4  kathir     cs    A+     90
std5    rani     ds    A+     90
std6   thara   csai     A     99
```

In [ ]:

```
#Column Access by Bracket Notation and Dot Notation
print(df['marks'])
print(df.course)
print(df[['grade','marks']])

#shape attribute
print(df.shape)

#Find Number Of Rows
print(df.shape[0])

#Find Number Of Columns
print(df.shape[1])

#Dtype Attribute
```

```
#Used to check Datatype of all columns
print(df.dtypes)

#Info
print(df.info())

print(df.describe())

#Select Specific Rows
print(df.loc["std3"])

#Update Value
df.loc['std1','grade']='B++'
print(df)
df.loc['std1','name']='Surya'
print(df)

#Delete Column
df.drop('grade',axis=1,inplace=True)
print(df)

#Delete Row
df.drop('std6',axis=0,inplace=True)
print(df)

#Sort Values
df.sort_values('marks',inplace=True)
print(df)

#Decending
df.sort_values('marks',ascending = False, inplace=True)
print(df)

#Statistical Function for particular Column
print(df['marks'].max())
print(df['marks'].min())
print(df['marks'].mean())
print(df['marks'].median())
print(df['marks'].std())
print(df['marks'].mode())
print(df['marks'].var())
```

In [ ]:

In [ ]: